NATIONAL UNIVERSITY OF TECHNOLOGY
(Department of Computer Science)

Course Title	:	Cloud Computing Lab	**Course code**	:	CS471
Batch	:	2023			
FM Name	:	Memona Ijaz	**Semester**	:	Spring 2026

Lab – 03
Data Cleaning & Analysis in Google Sheets

Objective: Clean raw sales data, extract unique insights, and build an advanced interactive dashboard.

Part 1: Data Cleaning (Standardizing & Formatting)

When you import the data.csv, you will notice inconsistencies (e.g., "North" vs "north", different date formats).

1.1 Standardizing Text (Casing)
The "Region" and "Category" columns have mixed casing (e.g., SOUTH, south, South).

1. Step: Insert two new columns next to "Region" and "Category".
2. Formula: In the new Region column, use: =PROPER(TRIM(C2))
3. Action: Drag this down for all 50 rows. Repeat for the "Category" column.
4. Finalize: Copy the new values and use Paste Special > Values Only over the original columns to replace the messy data.

1.2 Fixing Date Formats
The "Date" column contains multiple formats (e.g., 1 Feb 2025, 04-01-25, 03/01/2025).

1. Step: Select the entire "Date" column (Column B).
2. Action: Go to Format > Number > Date.
3. Result: Google Sheets will attempt to parse all entries into a uniform MM/DD/YYYY or DD/MM/YYYY format.

1.3 Handling Missing Values
In row 22 (Order ID 1022), the Unit Price for "Sugar" is missing.

1. Step: Find the row for Order 1022.
2. Action: Refer to other "Sugar" entries (e.g., Order 1010) where the price is 1700. Manually input 1700 into the empty cell to ensure your calculations are accurate.

1.4 Adding a Calculated Column
To analyze revenue, we need a "Total Sales" column.

1. Step: Create a new column "H" titled Total Sales.
2. Formula: In cell H2, type: =$G2 * $H2 (Quantity $\times$ Unit Price).
3. Action: Double-click the fill handle to apply to all rows.

A	B	C	D	E	F	G	H	I
Order ID	Date	Region	Salesperson	Category	Product	Quantity	Unit Price	Total Sales
1001	1/1/2025	North	Ali	Electronics	Mouse	5	1200	$6,000.00
1002	2/1/2025	South	Sara	Clothing	Jacket	2	4500	$9,000.00
1003	2/1/2025	West	Ahmed	Electronics	Keyboard	4	1800	$7,200.00
1004	3/1/2025	East	Fatima	Groceries	Rice	8	2200	$17,600.00
1005	4/1/2025	North	Bilal	Electronics	Headphones	6	2500	$15,000.00
1006	5/1/2025	South	Ayesha	Clothing	Shoes	3	5000	$15,000.00
1007	6/1/2025	West	Usman	Groceries	Oil	7	2100	$14,700.00
1008	7/1/2025	East	Zainab	Electronics	Speaker	2	6000	$12,000.00
1009	8/1/2025	North	Ali	Clothing	T-Shirt	10	1500	$15,000.00

Part 2: Extracting Unique Values

Before building a dashboard, it is helpful to know exactly how many unique Salespersons or Categories exist.

2.1 The UNIQUE Function

1. Step: Go to a new blank sheet or an empty area on the right.
2. Formula: =UNIQUE(E2:E51)
3. Result: This will list every unique category (Electronics, Clothing, Groceries) without duplicates.

2.2 Data Validation (Dropdowns)

1. Step: Select a blank cell (e.g., K2) and label it "Select Salesperson".
2. Action: Go to Insert > Dropdown (or Data > Data Validation).
3. Criteria: Select "Dropdown from a range" and select the unique Salesperson list you generated earlier. This will be used later for interactive filtering.

Unique Values		
Region	**Salesperson**	**Category**
North	Ali	Electronics
South	Sara	Clothing
West	Ahmed	Groceries
East	Fatima	
	Bilal	
	Ayesha	
	Usman	
	Zainab	

Part 3: Advanced Pivot Tables

Pivot tables allow us to summarize the 50 rows of data instantly.

3.1 Creating the Pivot Table

1. **Step:** Select the entire dataset (A1:I51).
2. **Action:** Go to **Insert > Pivot Table**. Choose "New Sheet".

<u>Pivot Table 1: Sales by Region</u>

1. Select full dataset
2. Click Insert ⟶ Pivot table
3. New Sheet

Set:

Rows → Region
Values → Sum of Total Sales
Values → Sum of Quantity
Format as currency.

Pivot Table 2: Product by Category

Pivot:
Rows → Category
Values → COUNT of Product
Chart Type → Column Chart
Shows product distribution.

Pivot Table 3: Sales by Category & Region
Rows → Category
Columns → Region
Values → Sum of Total Sales
This gives cross-analysis.

Pivot Table 4: Salesperson Performance
Rows → Salesperson
Values →
- Sum of Total Sales
- Sum of Quantity
- Average Unit Price

Sort by Total Sales (Descending)

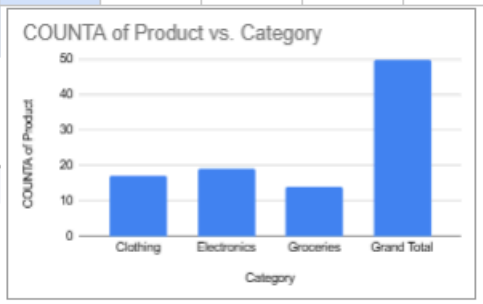
Part 4: Dashboard Creation

A dashboard is a visual summary of your cleaned data.
Chart 1: Sales by Region
1. Select Pivot Table 1
2. Click Insert → Chart
3. Choose Column Chart

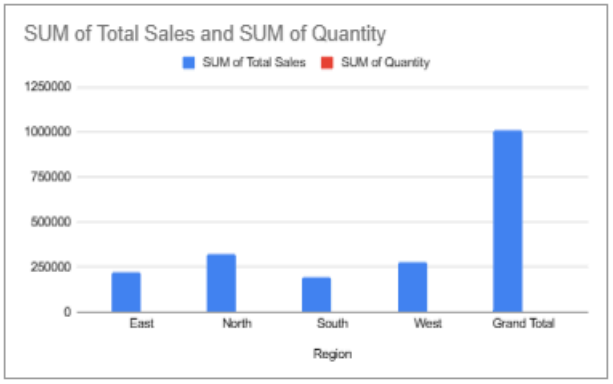
Chart 2: Category Sales Distribution

Use Pie Chart
Source: Category Pivot

Category	COUNTA of Product
Clothing	17
Electronics	19
Groceries	14
Grand Total	**50**



Region	SUM of Total Sales	SUM of Quantity
East	$219,100.00	55
North	$321,500.00	89
South	$193,000.00	65
West	$274,400.00	49
Grand Total	**$1,008,000.00**	**258**



PART 5 – SaaS Collaboration Task

Students must:

- Share sheet with 2 classmates
- One student adds comment
- One student edits formatting
- View Version History

Part 6 -Analytical Questions

1. Which region generated highest revenue?
2. Which category is most profitable?
3. Who is the best salesperson?
4. Which product sold the most?

Deliverables

Students must submit:

- Google Sheet link
- Screenshot of Pivot Tables
- Screenshot of Charts
- Screenshot of Dashboard